Isadora Santos

AI & Backend Engineer

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PROFESSIONAL SUMMARY

AI & Backend Engineer with 5+ years of enterprise experience, currently specializing in agentic AI workflows, backend modernization, and secure pipeline automation. Proven expertise in building reliable, enterprise-level production AI Agents, adaptive memory systems, and scalable infrastructure to automate complex operational tasks with rigorous software engineering practices.

TECHNICAL SKILLS

- **AI & Machine Learning:** Python, LangChain, LangGraph, LangSmith, ChromaDB, PostgreSQL pgvector, OpenAI API, Gemini API, Claude.
- **Engineering & MLOps:** Docker, Kubernetes (AKS/EKS), MLflow, Apache Airflow, FastAPI.
- **Backend & Databases:** Python, Java, SQL, JavaScript.
- **Cloud & Security:** AWS, Azure, SecOps, HashiCorp Vault, REST APIs, Microservices.

EDUCATION

- **Microsoft Generative AI Engineering |** Microsoft | Present
- **MLOps - Machine Learning Operations - Specialization** | Duke University | 2026
- **Artificial Intelligence Engineering - Masters** | GRAN University | 2025
- **Systems Analysis and Development Diploma** | UNINASSAU | 2022 - 2024
- **Computer Programming Diploma** | Georgian College | 2020 - 2022

FEATURED AI APPLICATIONS & SecOps PROJECTS

Stateful AI Vulnerability Agent (Adaptive Memory System)

- Architected an open-source agentic workflow using LangGraph and the Google Gemini API to automate complex triage and decision-making tasks, inspired by enterprise vulnerability management systems.
- Engineered a containerized FastAPI and Docker backend featuring a ChromaDB Retrieval-Augmented Generation (RAG) system, utilizing PostgreSQL pgvector for reliable state memory across expert feedback loops.

Secure MLOps Pipeline

- Developed a secure CI/CD MLOps pipeline to manage anomaly detection models, integrating continuous code quality checks (flake8) and automated security scanning (Trivy) via GitHub Actions.
- Established consistent model tracking and pipeline execution by containerizing the training environment and utilizing enterprise deployment standards like MLflow and LangSmith.

Agentic Data Extraction & Self-Healing ETL

- Built an intelligent, LangGraph-driven ETL pipeline capable of ingesting unstructured, messy data inputs and dynamically mapping them into strict analytical schemas.
- Developed an interactive React frontend designed for human-in-the-loop validation, allowing non-technical specialists to seamlessly review and correct the agent's extraction logic, backed by a PostgreSQL data persistence layer.

WORK EXPERIENCE

RBC - ROYAL BANK OF CANADA | Capital Markets - Quantitative Technology Services team

Senior Engineer | Montreal, QC | Apr 2024 - Apr 2025

- Architected a Java Spring Boot and Azure OpenAI AI Gateway serving 10+ distinct teams, acting as the integration layer between core infrastructure and LLM capabilities.
- Led the legacy migration of monolithic settlement systems to Azure Kubernetes Service (AKS) microservices, managing complex integrations with highly specialized financial backend services.
- Designed a secure Jenkins and MLflow continuous training pipeline for risk models enforcing strict Role-Based Access Control (RBAC), establishing automated feedback loops and reproducible deployment standards.

RBC - ROYAL BANK OF CANADA |Technology & Operations - Global Cybersecurity team

SecOps Engineer / DevSecOps | Toronto, ON | Jan 2022 - Apr 2024

- Architected and built a vulnerability management RAG assistant, automating threat triage and ticket generation architecture.
- Built an Apache Airflow, SQL, and AWS Lambda orchestration layer that automated enterprise vulnerability management, cutting Mean Time to Remediate (MTTR) by 60%.
- Developed 'Self-Healing' Python infrastructure bots and an internal Cortex XSOAR cloud notification system, automatically correcting misconfigured cloud resources and reducing incident response times by 60%.

MUNICIPALITY OF YORK REGION - Paramedics and Senior Services - Data & Continuous Quality Improvement team

Statistical Data Analyst | Newmarket, ON | Jan 2021 - May 2021

- Developed automated Python data pipelines and secure ETL processes for sensitive health data (PHI), ensuring strict compliance with privacy regulations.
- Engineered robust data extraction workflows to dynamically map complex, unstructured public health information into reliable schemas for reporting and analysis.
- Engineered a COVID-19 screening automation application using Microsoft PowerApps, SharePoint, and Power BI, managing the agile lifecycle and reducing manual errors by 70%.

LANGUAGES

- English: Fluent
- Portuguese: Fluent/Native
- Spanish: Advanced
- Russian: Intermediate
- French: Basic (currently learning)